

Co2 AT1

CSAD613 - DESIGN AND ANALYSIS
FOR ALGORITHM

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BTech(IT)

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85%

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28) Convex Hull - Outer Boundary Survey points. A land survey provides points $\{(0,1), (2,4), (5,5), (6,2), (4,0), (2,1)\}$. Apply the Brute-Force Convex Hull algorithm to determine the outer Boundary. Clearly interpret the problem and explain the need for convex hull computation. Check all possible line pairs and identify the Valid.....

Answer:

1. Introduction

The convex Hull of a set of points is the smallest convex polygon that encloses all the given points. It can be imagined as stretching a rubber band around the outermost points; when released the rubber band forms the convex hull.

2. Problem statement (understanding).

- $A = (0,1)$
- $B = (2,4)$
- $C = (5,5)$
- $D = (6,2)$
- $E = (4,0)$
- $F = (2,1)$

The objective is to determine the outer boundary using the Brute Force convex Hull Algorithm.

A Convex Hull is the smallest convex polygon that contains all the given points. It is like stretching a rubber band around all the survey points.

3. Need for Convex Hull Computation

- Eliminates interior points
- Helps estimate land area
- simplifies GIS and mapping
- Assists in property boundary identification.

4. Method Selection

Algorithm Used:

Brute Force Convex Hull Algorithm

Reason:

The problem specifically asks to:

- check every possible pair of points.
- Verify whether all remaining points lie on one side of the line
- If yes, that pair is a convex Hull edge.

5. Step by step solution

Step 1: plot the points
points coordinates
A (0,0)

B (2,4)
 C (5,5)
 D (6,2)
 E (4,0)
 F (2,1)

point F(2,1) lies inside boundary.

Step 2: Total possible line Pairs.

There are 6 points

Number of pairs

$$= {}^6C_2$$

$$= (6 \times 5) / 2$$

$$= 15 \text{ pairs}$$

The pairs are: AB, AC, AD, AE, AF, BC, BD, BE, BF, CD, CE, CF, DE, DF, EF

Step 3: Check Every pair

pair	Hull Edge
AB	Yes
AC	No
AD	No
AE	No
AF	No
BC	Yes
BD	No
BE	No
BF	No

CD	Yes
CE	No
DE	No
DF	No
EF	Yes

Step 4: Excluding Interior point

$$F = (2, 1)$$

F is excluded from the convex hull

Step 5: Valid Hull Edges

The convex hull consists of:

- AB
- BC
- CD
- DE
- EA

Step 6: Convex Hull Vertices

Moving in clockwise order:

$$(0, 1) \rightarrow (2, 4) \rightarrow (5, 5) \rightarrow (6, 2) \rightarrow (4, 0) \rightarrow (0, 1)$$

Interior point:

$$(2, 1)$$

5. calculations

No. of point pairs

$$\binom{6}{2} = 15$$

remaining point checked

$$= 6 - 2$$

$$= 4 \text{ points}$$

Total comparisons

$$= 15 \times 4$$

$$= 60 \text{ comparisons}$$

Time complexity

Number of pairs

$$= O(n^2)$$

= Each pair checks remaining points

$$= O(n)$$

Overall Time complexity

$$= O(n^3)$$

Space complexity:

$$O(1)$$

Final Answer

Conve Hull Vertices = $(0, 1), (2, 4), (5, 5), (6, 2), (4, 0)$

Interior Point
(2,1)

Total line pairs
15

Total comparisons
60

Time complexity
 $O(n^3)$